

Amber Driver Recovery

Issue Description:

Display freeze issue was observed in BAIC project. The display remained freeze and any of the screen updates were not happening.

Root Cause:

It is observed that the amber driver sync object is returning `MML_ERR_GDC_SYNC_TIMEOUT` always when the issue happens. As per Project team, it has been observed only once and can't be reproduced later.

Solution:

In order to support such a scenario it has been decided to implement some recovery strategy from graphics manager side as a possible solution. A recovery mechanism need to be done while display remains freeze.

Hence the Amber Driver need to be restarted along with proper restart of graphics manager 02 and also ensure that the current active screens and animation element values are undisturbed.

Configuration to be done in hmi_gfx_mgr02.cfg

Enable below macro if Amber Driver restart(Recovery Mechanism) is required during run time.

```
#define GFX_MGR02_AMBER_DRIVER_RUNTIME_RECOVERY
```

Configuration to restart the amber driver when it's in BUSY state for prolonged duration. Duration indicating how long the driver status shall be busy. Configured as 5 minutes = (60 * 5 = 300)

```
#define AMBER_DRIVER_BUSY_STATUS_DURATION_IN_SECS          300
```

The CmpActive routine of GdtCdd is used to monitor the Amber Driver status.

The scheduling period of CmpActive routine need to be configured here.

The schedule duration of GDT Cdd module in Da Vinci configuration need to be configured in below macro

E.g.: Amber Driver status monitored once in every 16 milliseconds through CmpActive routine.

```
#define AMBER_DRIVER_STATUS_EVALUATION_FREQ_IN_MILLISECS    16
```

Please Note: The maximum duration can be **960** seconds (16 minutes) for an evaluation frequency of **16 milliseconds**. When the evaluation frequency is 8 milliseconds then maximum 8 minutes duration can be given.

Known Limitations

During recovering phase display will go blank and then redisplay the current active screen. This may appear to be flicker/red flash for a moment which cannot be avoided.